ANAGHA S K

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Summary

AI Engineer with hands-on experience building computer vision systems for industrial safety, from problem definition and dataset creation to model training, evaluation and multi-camera optimization. Core team member of AEGIS at Techcrafter, spanning object detection, pose estimation and Transformer-based NLP. B.E. in Computer Science (AI & ML, CGPA 8.9)

Experience

Techcrafter **Apr 2026 – Present**
AI Engineer (Intern → Full-time) *Bengaluru*

- Part of the core team behind AEGIS, an AI-based industrial safety platform. Researched the problem, defined the problem statement, and helped shape the product from scratch.
- Took part in every stage of the AI work, from data collection and annotation through model research, training, evaluation and deployment-readiness.
- Compared and benchmarked state-of-the-art computer vision models on real factory camera conditions, and chose the best-performing approach using measured results.
- Improved speed and reliability on limited hardware, raising multi-camera throughput by up to nearly 4x.
- Worked with the CEO and team in regular progress reviews to turn research findings into product decisions.

Hindustan Aeronautics Limited (HAL) **Apr 2025 – May 2025**
Project Intern *Bengaluru*

- Built an offline English-to-Hindi/Kannada translation desktop app for internal use, with all text kept local.
- Selected pretrained Transformer models from Hugging Face for each language: MarianMT (Helsinki-NLP OPUS-MT) for Hindi and Meta's NLLB-200 (600M-parameter distilled) for Kannada.
- Implemented the translation pipeline in Python and PyTorch, covering tokenization and beam-search decoding, with input validation and error handling.
- Designed a Tkinter interface with language selection and proper Devanagari and Kannada fonts, for easy use.

Projects

AEGIS: AI-Enabled Guardian for Industrial Safety | *Python, OpenCV, PyTorch* **Apr 2026 – Present**

- AI system that monitors factory CCTV for missing PPE, falls and unsafe worker postures, tested on **8 concurrent camera streams** on a single 8 GB laptop GPU.
- Built fall detection and unsafe-posture analysis (**5 ergonomic risk types**) using pose estimation, raising per-camera throughput from **2.5 to 9.6 fps** for falls and **2.4 to 4.4 fps** for posture.
- PPE detection: collected the dataset from elevated top-angle CCTV views, annotated and augmented it, growing it from **3.5k to ∼7k images across 9 classes**, which lifted worker detection from a **60–69% plateau to 77.3% mAP@50** (precision 0.77, recall 0.78), after benchmarking multiple model versions.
- Validated with **1,100+ automated tests** covering conditions such as occlusion, camera drops and restarts.

NotifyNPick: Real-Time Inventory Management | *Python, OpenCV, Tesseract, Tkinter* | Demo Link

- Led launch of an automated inventory system for Village Mail & More (San Diego), cutting manual labor.
- Automated ticketing and tracking with OCR and barcode recognition (OpenCV, PyZbar, Tesseract).
- Integrated Dropbox and email APIs for real-time inventory updates and notifications.

HireMate AI: Career & Interview Prep | *Python, Flask, LLM APIs, Web Speech API*

- Developing an AI placement-prep platform with ATS-compliant resume generation and job–resume fit scoring.
- Implemented AI mock interviews and adaptive aptitude tests with speech-based evaluation of clarity and confidence.
- Designed an AI-simulated group discussion module with real-time feedback and communication analytics.

Technical Skills

Languages: Python, SQL
Computer Vision: Object Detection, Pose Estimation, OpenCV, Ultralytics, Roboflow, Data Augmentation, OCR
ML & NLP: PyTorch, Hugging Face Transformers, Scikit-learn, NumPy, Pandas, Model Fine-tuning & Evaluation
Tools: Git, GitHub, Flask, Tkinter, LLM APIs, Power BI

Education

JSS Academy of Technical Education, Bengaluru **2022 – 2026**
Bachelor of Engineering in Computer Science (Artificial Intelligence and Machine Learning) *CGPA: 8.9/10*

Indian Institute of Technology, Mandi **2025 – 2026**
Minor in Artificial Intelligence

Leadership / Extracurricular

Arthouse (Design Club) – *Designer* JSSATEB

Meraki (Drama Club) – *Technical Director & Scriptwriter* JSSATEB

NSS – *Social Service Volunteer* JSSATEB